

Trading bot master specification

Build one working, measurable paper-trading system first. Preserve the path to fully autonomous live execution.

Prepared for Frank from his revised 11-page bot audit (29 September 2026), seven photos of the Quant Farming Funding Account Risk Management Guide, his accumulation screenshot, and his clarifications in this conversation. This is a consolidated implementation brief; it does not certify completed code changes or profitable future results.

The operating model Frank clarified

Stage	Who decides / who executes
1. Autonomous paper	The bot detects setups, sizes positions, simulates entries and exits, manages risk, and records outcomes without Frank placing each simulated trade.
2. Prop validation	The bot provides entry, management and exit alerts. Frank personally reviews and places the orders on the prop platform. The bot has no prop order-entry, modification or cancellation capability.
3. Personal live	The same validated decision and risk logic can execute autonomously through a permitted broker connection after the promotion gates are met and Frank explicitly enables that mode.

Performance context: Frank identifies the photographed guide as results from a proven successful live signal engine. Its reported 69.9% win rate is a source-engine benchmark. Frank's current bot is a separate implementation; its audit reports zero executed strategy paper trades. Preserve both facts without treating the source performance as already reproduced.

Read first	Purpose
Pages 2-6	Decisions, current baseline, guide integration, corrected math and sizing.
Pages 7-13	Strategy rules, regimes, accumulation, execution safety, manual prop workflow and rules.
Pages 14-17	Evidence, promotion criteria, data contracts and implementation sequence.
Pages 18-20	Defaults, remaining decisions, Claude Code instructions and sources.
Standalone edition	The implementation specification only; source audit and photos are not appended.

Authority: Frank's latest explicit instructions control. Pages 1-19 reconcile the source documents; conflicting legacy wording in the source materials is historical reference, not an instruction to bypass this specification. Proposed defaults are labeled and must never be represented as proven source-engine settings.

Decisions carried into the build

Original answer / intent	Consolidated requirement
Fully autonomous bot	The final product is autonomous. Autonomous paper execution comes first; prop mode remains advisory with manual execution.
70-75% win-rate objective	Target at least 70%; 75% is the higher objective. Always display observed performance and uncertainty separately from target and source-engine results.
US 30, VIX, NAS1, VXI, XAUL, ES, MES	Retain this as the desired research universe. Proposed first paper instrument: MES, already present in Frank's answer. Resolve exact exchange symbols and contract metadata before enabling any additional market.
Apex / Tradovate	Retain as the intended manual prop-validation route. Do not replace it with Topstep simply because the guide discusses Topstep.
Forex Factory	Use for event-calendar context, subject to reliable access. It is not a complete specification of the futures price/execution feed. Reuse the functioning futures provider in the repository where possible.
Multi-timeframe liquidity strategy	Preserve daily bias, H4/H1 liquidity, reversal and continuation setups, lower-timeframe confirmation, structural stop and liquidity target. Formalize them in pages 7-8.
Cash-open entry / no news-direction trades	The revised answer removes the old first-30-minute wait and says enter after the market opens. Proposed MES pilot: 09:30-11:30 America/New_York cash-open window [W10], subject to completed-bar, spread, volatility and event checks. No premarket entry.
0.25-0.5% per trade; 1% daily; 25% drawdown allowance	These are Frank's revised nominal-capital and daily ceilings, not firm rules. Apply the smallest allowed dollar limit after usable-loss-capacity checks. Proposed paper pilot remains \$70 maximum initial trade budget and \$150 net daily stop.
1-3 micro contracts	Revised answer is an upper size range, not an obligation to trade. Start at one MES micro; allow up to three only under a later validated, versioned profile and all dollar-risk, firm and margin caps. Zero contracts means skip.
100+ trades, 70%, 4+ weeks	Retain as the minimum initial evidence floor. Add net expectancy, execution integrity, rule compliance and uncertainty reporting; four weeks alone cannot pass.
100+ trades, 60%, two payouts before live	Retain two payouts and minimum initial live size. Do not silently lower the stated 70% objective to 60%; a different target requires an explicit strategy-policy revision.
Eight days / five \$50+ days / 30% rules	Move into the applicable legacy funded/payout profile. They are neither a daily loss limit nor a universal evaluation pass criterion.

The revised audit now fills both cells: one to three micros and exit at stop or target. The one-micro pilot and protective time/session exits are explicitly labeled implementation defaults, not quotations from Frank. The 25% daily drawdown phrase is ambiguous; page 6 operationalizes it conservatively until the exact account definition is known.

Current bot: retain, connect, verify

Treat the supplied audit as the current project handoff, as Frank requested. Confirm its findings against the current checkout during implementation, but do not commission another broad research program before building the paper execution path.

Audit-reported state	Implementation consequence
8,699 decisions; zero strategy trades; accounts at \$50,000	Current local-strategy win rate and expectancy are undefined. Instrument the complete path and collect actual simulated fills.
Adaptive lane hard-coded comparison-only / paper not permitted	Replace with an explicit execution mode. Allow the autonomous paper adapter while keeping real-order adapters unavailable.
Candidate reports never reach paper engine	Connect candidate -> deterministic strategy confirmation -> portfolio/risk check -> simulated order -> fills -> ledger.
Baseline always reports no signals	Remove or clearly disable the placeholder from the active path; do not count its inactivity as strategy evidence.
294 candidates lacked sweep; 136 also lacked displacement	Inspect causal feature calculations and definitions. A rejected candidate is not a trade; do not relax confirmations or force entries merely to increase count.
Risk engine, kill switch and ledger have component evidence	Reuse them. Verify end-to-end accounting, recovery and exit behavior; component tests do not prove broker readiness.
Hyperliquid, NinjaTrader and IBKR read data; Rithmic placeholder	Do not invent an existing order adapter. Prop alerts need no trading adapter. A later personal-live adapter is a separate deliverable.
1.5% daily loss / 3% drawdown in code	Preserve as outer internal caps until semantics are checked. Add stricter pilot limits and exact account-specific rules; never broaden them to make trades pass.
Short Linux tests; no sustained Windows/broker trial	Validate the selected Windows startup and full-session paper path. Classify platform-dependent test failures explicitly.

Setup work limited to the chosen path

Select the existing active branch/checkout without discarding other work. Repair candle-window recording if used; add one documented startup command; resolve required dependencies and tracked synthetic fixtures; replace user-specific paths. Preserve credentials outside Git. Fix Docker only if it is the deployment path. Preserve checksum-sensitive files and the project's line-ending rules.

Correct the evidence ledger

The audit totals 5,479 passing + 112 failing tests = 5,591, not 5,592: classify the remaining result. Its 2,243 baseline + 6,162 adaptive counts leave 294 decisions; document whether candidates are a separate bucket or a subset. Distinguish synthetic ledger tests from strategy trades. These are bookkeeping fixes, not reasons to suspend paper development.

Guide parameters: what becomes policy

Photographed parameter	Use in Frank's specification
69.9% live-engine win rate; 4-5 trades/week	Record as user-attested source-engine results [P1]. Request the source ledger/version if available, but do not block basic paper implementation on it. Use trade frequency for planning, never as a quota.
65-72% planning range; 4 trades/week; 1:1 payoff	Retain as scenario assumptions [P1], not confidence bounds or guarantees. Add adverse 50%, 55% and 60% scenarios and empirically clustered losses.
Risk-to-reward 1.0721:1	Retain the printed value as ambiguous [P1]. Literal reward/risk is 0.93275; reversed notation gives 1.0721. Do not choose the more profitable meaning silently.
0.75-1% fixed risk	Keep as a research comparison [P2], disabled for default prop/paper sizing. Budget in dollars using remaining loss capacity, fees and structural stop.
Increase to 1.25-1.5% after +5%	Optional future experiment [P2], disabled initially. Profits do not necessarily expand usable cushion under a trailing floor. No automatic growth escalation.
1.5-2% fast-pass; 2-3 week passage	Do not adopt as production policy [P3-P4]. Higher risk may accelerate wins and failures; pass probability and expected total costs matter. No account-reset farming.
Reduce to 0.5-1% once funded	Adopt the principle of protecting funded capital [P4], not the percentage. Use the stricter applicable dollar-risk limits at every stage.
Clean trend, momentum, follow-through	Translate into causal, setup-specific filters [P5]. Trends may favor continuation and undermine a reversal; test each separately.
Reduce risk after clustered losses	Use deterministic pause/review rules [P5-P6]. Never increase size after a loss or average into losers.
Contract sizing / 1% of \$100k = \$1,000	The arithmetic is correct; the printed sizing expression is not [P6]. Replace it with the division-and-floor formula on page 6.
2-3 losses; 4-6 trade drawdowns are normal	Possible, not upper bounds [P6]. Model longer sequences, drawdown depth, duration and regime persistence.
Perfect execution / full compliance / patience	Model imperfect fills and operational failures [P7]. Discipline supports a valid edge; it cannot guarantee profits. Qualitative safety rankings are not statistical risk estimates.

The seven images do not supply the source engine's complete signal code, sample size, trade dates or full intervening guide pages. Do not claim an exact clone of its performance. The photographs improve risk planning; Frank's completed strategy remains the signal specification.

Corrected expectancy and timing

Define one completed position, from flat to flat, as one trade. Aggregate scale-outs into that position. A win has positive net P&L after trading costs. Zero net outcomes remain in the denominator and are reported separately. Decisions, alerts, copied accounts and order fills are not independent trades.

R dollars = original planned stop loss before costs; never rebase R after moving a stop.

Net expectancy in R = mean of (each position's net P&L / its initial R).

For binary gross outcomes: $E/R = p \times b - (1-p) \times a - c$.

p = win probability; b = mean winner/R; a = mean loser/R; c = mean trading cost/R.

Win rate / payoff	Gross expectancy	Four trades at 1% risk
65%, +1R / -1R	+0.30R	+1.20% per week
70%, +1R / -1R	+0.40R	+1.60% per week
72%, +1R / -1R	+0.44R	+1.76% per week
75%, +1R / -1R	+0.50R	+2.00% per week
70%, +0.20R / -1R	-0.16R	Negative before costs

For $a = 1$, break-even $p = (1+c)/(1+b)$. Risking 5 to target 1 means $b = 0.2$ and needs more than 83.33% wins before costs. A firm's maximum allowed risk/reward ratio is not an optimal payoff. At the reported 69.9%, the two interpretations of 1.0721:1 imply about +0.351R or +0.448R gross.

Target / source estimate	Arithmetic weeks at 65% / 72%
10% / about 8-9 or 5-6 weeks	8.33 / 5.68
8% first phase	6.67 / 4.55
5% second phase	4.17 / 2.84
Separate 8% + 5% phases	10.83 / 7.39

These divisions assume four trades weekly, fixed starting-balance risk, 1:1 realized outcomes and zero costs. At 0.75% risk the weekly range becomes 0.90-1.32%. For the +5% scale-up model, simple 10% target arithmetic is about 6.94-7.50 weeks at 65%, or 4.73-5.11 at 72%. Thus 4-6 weeks is not supported throughout the source range.

At 70% wins and 2% risk, four 1:1 trades imply +3.2% gross weekly; at 1.5% risk, +2.4%. Strong-condition filters may reduce frequency. Replace target/mean-return forecasts with probability of pass before breach, time conditional on passing, unresolved paths, and total fees across attempts. Fixed account fees belong in program economics, not a fictitious per-fill commission.

Risk budget and futures sizing

```
stop_ticks = abs(entry_price - stop_price) / tick_size
unit_loss = stop_ticks x tick_value + round_trip_cost + stressed_slippage
contracts = floor(available_trade_budget / unit_loss)
final_quantity = min(contracts, internal_cap, firm_cap, margin_cap)
```

Round prices to valid ticks without understating risk. Use exact contract metadata and integer quantity. If quantity is zero, skip. Do not tighten a structural stop to force affordability. A stop order limits intended risk but cannot guarantee its execution price during gaps or outages.

Instrument [W2]	Point value	Tick size / value	20-tick loss before costs
MES	\$5	0.25 / \$1.25	\$25 per contract
MNQ	\$2	0.25 / \$0.50	\$10 per contract

Example only: with a \$100 trade budget, a 20-tick MES stop and assumed \$3 all-in cost/slippage allowance per contract, $\text{floor}(100/28) = 3$ contracts before other caps. The one-micro pilot cap reduces this to one. Fees and slippage must be replaced with the chosen platform's measured or scheduled values.

Account capacity is a state, not the advertised balance

Track liquidation equity, the active loss floor, floor-update method, daily starting equity, open risk, pending-order risk and withdrawal effects. Compute the smallest projected slack to every applicable limit after all positions hit stressed exits. For a floor based on unrealized highs, update it along the simulated path before judging survival.

C = remaining usable loss capacity after existing exposure and reserve
 $\text{new_trade_budget} = \max(0, \min(\text{pilot_dollar_cap}, 0.05 \times C, \text{remaining_daily_budget}, \text{remaining_portfolio_budget}, \text{firm_trade_limit}))$

The 5% coefficient is a **proposed cautious paper default**, not a guide parameter or optimized result. New orders reserve capacity atomically. Never count unrealized profit twice as both cushion and committed risk. In multi-instrument work, aggregate correlated exposure; the initial pilot allows only one position.

Proposed isolated paper profile

Start simulated equity at \$50,000. Use a \$48,500 static synthetic floor, a \$100 execution reserve, a \$100 maximum new-trade budget, a \$150 net daily stop, one MES micro maximum and three entries per session maximum. These are stricter pilot settings, not Apex rules. Initial C before exposure is \$1,400, so the 5% capacity cap is \$70. Frank's revised 0.25-0.5% nominal trade range means \$125-\$250 and 1% nominal daily ceiling means \$500; the pilot caps remain smaller. His 25% of total drawdown per day is an internal proposed ceiling: with a \$1,500 synthetic allowance it is \$375, still above the \$150 pilot stop. Recompute these comparisons as actual headroom changes; never apply a nominal percentage instead of a tighter real loss floor.

No averaging down, martingale, doubling after losses, backward stop widening or risk escalation after gains. Pause new entries after three consecutive net losing positions until the next session and logged review. Continue protective exit handling while paused. Frank's three-micro limit is a future maximum, never a minimum quantity. A firm-profile simulator later replaces the synthetic floor semantics; it never silently inherits them.

Make the market structure computable

First inspect existing strategy definitions and retain equivalent behavior where it is already coherent and tested. Where the repository is silent, use the proposed definitions below in a clearly named pilot variant. These definitions are not supplied by the source engine and must not be advertised as its proven settings.

Concept	Proposed deterministic interpretation for paper
Causality / bars	Use completed exchange-session H4, H1, M15, M5 and M1 bars. Record bar start/end and availability time. Use only features available before the decision.
Confirmed swing	A high exceeds highs of two bars on each side; inverse for lows. A swing becomes usable only after the second right-hand bar closes. Equal prices are not strict pivots.
Bias	On H1 and H4, a close beyond the most recent confirmed swing high sets bullish structure; below swing low sets bearish. The latest such event persists until opposite break. Trade only when both agree; otherwise neutral.
FVG	Three closed bars A/B/C: bullish if C.low > A.high; bearish if C.high < A.low. Gap must span at least one tick. Log creation, bounds, touches, full fill and invalidation separately.
Inverse FVG	A close through the opposite boundary invalidates an existing gap; a later retest from the other side can qualify as an inverse gap only after confirmation in the intended direction.
Liquidity levels	Prior completed trading-day high/low; prior defined Asia/London session high/low; confirmed H1/H4 swing levels and unfilled H1/H4 gaps. Freeze the level set at setup creation.
Equal / stacked highs or lows	Proposed tolerance: two ticks between at least two confirmed levels. Keep level count and age; do not call a cluster observable before its pivots are confirmed.
Sweep / displacement	Sweep: excursion at least one tick beyond a marked level followed by a close back through it on the confirmation timeframe. Displacement: candle body $\geq 1.0 \times$ prior completed ATR(14), close in outer 25% toward trade direction.
BOS / CISD	BOS: close at least one tick beyond a confirmed structural swing. Pilot CISD: close through the open of the most recent opposite-direction candle preceding the displacement. Record candle IDs so the definition is reproducible.
Equilibrium / OTE	Equilibrium = midpoint of an explicitly identified leg or gap. If OTE is required, use a proposed 62-79% retracement of the last confirmed impulse, with endpoints frozen when setup forms.
Low-resistance liquidity	Select the nearest eligible pre-existing liquidity level in the intended direction with no closer opposing H1/H4 obstacle. Reject if distance fails the tested payoff requirement.

Unit tests must use hand-labeled synthetic examples of valid and invalid structures, including delayed confirmation. Avoid tests that simply repeat the production code. Refine an interpretation only by a versioned decision; never rewrite history to improve backtest outcomes.

Entry, exit and setup lifecycle

Setup	Required sequence
REVERSAL_R1	Aligned H4/H1 bias -> price reaches a premarked counter-side liquidity level -> sweep and rejection -> M15 or M5 inverse FVG or structural break in the bias direction, with displacement -> M1 confirming retest -> valid structural stop and directional liquidity target.
CONTINUATION_C1	Aligned H4/H1 bias -> unconsumed directional liquidity target remains ahead -> M5 displacement creates an aligned FVG -> qualifying pullback/OTE -> M1 gap respect and CISD in bias direction -> valid structural stop and target. A reversal-style sweep is not automatically required for this separately defined setup.

Frank's written continuation entry does not explicitly require a sweep. Do not fix the audit's missing-sweep issue by weakening every setup. Apply the confirmation matrix to the correct setup ID. Run both as separately logged paper variants, with at most one active position across them.

Proposed pilot order and exit rules

- **Entry:** after the confirming M1 candle closes, submit the simulated order at the next available executable quote. Cancel if the setup is invalidated, the target is consumed, or the entry is stale. No fill at the confirmation candle's already-known close.
- **Expiry:** proposed setup lifetime 15 minutes after lower-timeframe confirmation; executable entry intent lifetime 60 seconds. A price move of more than two ticks from the permitted entry zone invalidates the intent unless freshly recalculated.
- **Stop:** long below the relevant sweep/entry swing low, short above its high, with proposed one-tick buffer. A continuation uses the confirmed gap-respecting candle/structural invalidation as specified by its setup. Choose before entry and store the reference.
- **Target:** first qualifying premarked liquidity level in the trade direction. Require at least 1.0 gross reward/risk in the proposed pilot. For initial paper only, check scenario expectancy using $p = 0.65$, $a = 1$, proposed payoff and modeled costs; label it an assumption, not measured edge. No historical profit record is required to start paper. Do not move a nearer target further away to make the ratio pass.
- **Management:** Frank's revised ordinary exit is at the stop or take-profit. Keep fixed initial stop/target, with no partials or discretionary trailing in the first pilot. A proposed 30-minute maximum hold, scheduled-event flatten and session cutoff are additional protective exits; log each cause separately and validate their effect before changing the rule.
- **Concurrency:** one active setup commitment and position; when both qualify at the same time, skip the conflict and log both IDs in the pilot. One filled attempt per setup; no repeated entries from the same candle.

Lifecycle and reasons

OBSERVING -> ARMED -> CONFIRMED -> RISK_APPROVED -> ENTRY_PENDING
-> OPEN -> EXIT_PENDING -> CLOSED

Alternative terminal states: EXPIRED, INVALIDATED, REJECTED, CANCELLED.

Every transition carries a UTC timestamp, market-data timestamp, strategy/config version and reason code. Missing data, confirmation or metadata means no new entry. A valid no-trade session is acceptable; a broken route must instead raise an operational fault. Separate those outcomes in the report.

Regimes, hours and event handling

The guide's trend/momentum preference is useful context. It is not evidence that all of Frank's reversal and continuation patterns work in every trend. Pre-entry filters must use information then available, not eventual follow-through or a breakout later labeled false.

Dimension	Pilot policy / measurement
Trend, chop and accumulation	Calculate a 20-bar efficiency ratio on completed M5 closes: $\text{abs}(\text{net change}) / \text{sum}(\text{abs}(\text{each change}))$; zero denominator -> unknown. Proposed continuation eligibility ≥ 0.35 ; reversal evaluated separately. Track a distinct accumulation/compression candidate under the page 10 state rules. Thresholds are pilot hypotheses.
Volatility	Track M5 ATR(14) versus a trailing 60-session same-time distribution. Proposed entry band: 10th-90th percentile. Outside the band -> no new entry. Insufficient history -> log filter unavailable, do not fill an enabled filtered variant.
Liquidity / quotes	Require a valid non-crossed executable bid/ask. Proposed MES maximum spread: two ticks. Historical bar-only data is unsuitable for claiming validated spread or queue behavior.
Compression / false-break risk	Record range contraction, gap size/ATR, boundary tests, target distance, and failed breaks. The page 10 accumulation candidate prevents interior crossover-only entries; additional slope and volume tests remain diagnostics until validated.
Setup-specific reporting	Separate continuation/reversal, long/short, volatility buckets, session segment, spread and event proximity. If a bucket lacks evidence, limit the claim to observed regimes.

One paper session first; global levels remain available

The revised answer permits entry after the market opens and bars premarket entries; it no longer requires a 30-minute delay. Proposed first MES entry window: 09:30-11:30 America/New_York, with no entries at or after 11:30 and flat by 11:30. The US equity cash open is 09:30 ET [W10]; the form's 09:00 wording is not that cash open. Enter only after the required completed confirmation bar and valid executable quote, so do not force a 09:30 fill. Use UTC internally and zone-aware labels, not fixed EST or Phoenix time.

Proposed level-building windows: Asia 19:00-00:00 America/New_York and London 02:00-05:00 America/New_York. These are explicitly named strategy windows, not assertions about city exchange hours. Use only completed windows; preserve user alternatives as separate configs. Apply exchange holidays, shortened sessions, maintenance and contract rollover.

News policy

Do not derive trade direction from news. Proposed major scheduled-event blackout: no new entries 10 minutes before through 15 minutes after; flatten five minutes before if a position could overlap the event. Include relevant inflation, employment and central-bank events. Reopen only with fresh quotes and normal spread/volatility. An unavailable or stale required calendar disables new entries for the affected session; protective exits remain active.

The event windows are proposed operating defaults, not universal prop rules. Measure filtered versus unfiltered performance chronologically, including lost opportunities and net costs. Do not select thresholds on the final holdout.

Accumulation and compression candidate

Frank asked to factor in accumulation, and the supplied screenshot recommends a moving-average slope clutter filter and avoiding thin sessions. Price alone cannot establish that institutions are accumulating; classify an observable compression or accumulation **candidate**, then test whether it improves decisions. The following thresholds are proposed paper interpretations, not proven source-engine settings.

State / measurement	Proposed causal pilot treatment
Detect range	On each completed M5 bar, use the prior 21 completed M5 closes (20 return intervals) and their bars. Freeze the high, low, age and contract. Require prior $ATR(14) > 0$, range width / prior $ATR(14) \leq 3.0$, and 20-interval efficiency ratio ≤ 0.25 . Missing history or invalid values -> UNKNOWN.
Name honestly	Mark <code>RANGE_COMPRESSION_CANDIDATE</code> ; add <code>ACCUMULATION_CANDIDATE</code> only as a research label when prior downward context and valid volume/order-flow evidence are specified. Do not infer net buying, absorption or an upward breakout from a tight price range alone. A symmetric distribution candidate is possible.
Inside the range	Do not enter from an RSI extreme or moving-average crossover alone. Keep <code>REVERSAL_R1</code> eligible only if its premarked boundary sweep, bias, M15/M5 and M1 confirmation, structural stop, usable target and risk checks all pass; log range-edge versus interior opportunities separately.
Release / breakout	A completed M5 close beyond a frozen boundary begins <code>TRANSITION</code> , not an immediate filled breakout. <code>CONTINUATION_C1</code> still needs its aligned H4/H1 bias, displacement/FVG, pullback, M1 confirmation, acceptable target and standard regime, cost and risk checks. A failed break re-entering the range is recorded as such.
Clutter / thin market	Log normalized moving-average slope, crossover count, comparable contract volume, spread, quote age, and time-of-day liquidity. A flat slope or thin period can support <code>WAIT/ABSTAIN</code> under the existing data and session rules. Do not add an untested numerical slope gate or invent order-book evidence from OHLC bars.
Lifecycle	Re-evaluate only after bars close; never relabel past entries from the later breakout. Expire stale ranges, freeze boundaries at setup creation, and retain the protective stop/target/time rules for an already-open position. Record entry-eligible, skipped and unknown states with reason codes.

How to decide whether the filter helps

Compare the unchanged pilot against an accumulation-aware variant on chronological forward data. Report missed opportunities, trade count, net expectancy, win rate, false-break rate, slippage, drawdown and sample size by setup and session. Keep the variant versioned; do not promote it because a chart looks convincing or because it raises the in-sample win rate.

The screenshot is a prompt for hypotheses about sideways crossovers and low liquidity [1], not measured performance evidence. Price-only compression is not proof of institutional inventory. No new live or prop route is authorized by this state classification.

Execution safety without endless gates

Failure or control	Required behavior
Mode isolation	PAPER_AUTO can use only a simulator. PROP_MANUAL_ALERTS can generate alerts and reconcile read-only records, never call a trade endpoint or manipulate the platform UI. LIVE_AUTO requires a separately enabled personal broker adapter.
Stale / invalid market data	Default new-entry quote age limit: two seconds on a live quote feed. Check timestamps, gaps and clock skew. No new entries on fault; existing protective orders are not cancelled merely because data is stale.
Duplicate decisions / retries	Use durable signal_id, intent_id and order_id. Replayed market events or restart retries cannot create a second order. Reserve risk before submission and release it only on confirmed cancellation or terminal completion.
Partial / rejected fills	Update quantity from actual fills. Protective coverage must match filled quantity. Rejected or unconfirmed protection triggers an emergency handling state and blocks new entries.
Restart / disconnect	Persist state; reconcile broker/simulator positions and working orders before resuming. Unknown account state blocks entries. Test open-position recovery and late fills after cancellation.
Daily stop / loss floor	Monitor realized plus unrealized net P&L and projected worst-case exit. Pause new entries before the external breach boundary. In automated modes, cancel entries and flatten through the approved path when the internal hard stop requires it.
Manual prop safety	Alert Frank to exit/cancel/flatten; do not execute it. Native protective orders must be configured by Frank at entry. Record acknowledgement and actual action time; unresolved status blocks further trade alerts.
Kill switch	Persistent across restart. In automated modes cancel pending entries, attempt controlled flattening and verify final state. In prop mode issue urgent instructions and require human reconciliation; do not falsely report automatic flattening.
Credentials / account identity	No credentials in reports or Git. Verify environment, account ID, instrument and permissions at startup. Real-money credentials are unnecessary for the initial paper path.
Runtime decision control	Entry/exit/risk decisions are deterministic and versioned. An LLM may explain logs or assist research; it may not override risk, invent prices, self-modify parameters or issue unaudited discretionary orders.

These controls replace the audit's blanket instruction to wait for a safety problem before adding a check. They are a finite known-failure test set. Once the paper path and its essential controls pass, run full sessions; do not keep creating new research gates that prevent data collection.

Simulator fill policy: market buys use ask, sells use bid plus calibrated latency/slippage; limit touch alone does not prove fill. Resolve stop/target same-bar ambiguity with finer data or conservative ordering. Model fees, partial fills, gaps and protective-order rejection.

Prop testing with Frank placing trades

The bot may autonomously analyze and alert. It must not interact with the prop order-submission interface.

The current Apex User Agreement permits signal/alert tools that do not place, modify or cancel orders, and analysis tools that do not interact with order submission. It also permits user-configured platform-native protective orders. This supports Frank's clarified manual workflow; it does not authorize an autonomous Apex executor. The agreement controls over conflicting guidance [W1].

Entry alert contract

Show signal ID; strategy version; instrument and expiry; long/short; setup; detection and expiry time; entry zone; current price; stop; target; suggested integer quantity; worst-case dollar risk; payoff estimate; active account-rule status; invalidation; and the instruction to skip if stale. A recommendation is not an order and is not logged as a fill.

The manual loop

- The bot produces a qualifying alert. Frank reviews the current market and personally enters the order and native protection on the prop platform.
- Frank confirms entered, skipped or expired. Import read-only fills when available, or record actual order/fill time, price, size, stop and target manually.
- Reconcile actual position before issuing management alerts. Alert clearly for exit, cancellation, time exit and emergency flattening. Frank executes each required action.
- If actual fill or protection differs materially, recalculate the advisory risk state and flag the deviation. Do not edit the platform or silently replace actual records with theoretical fills.
- Maintain a parallel autonomous paper position from the same signal. Compare theoretical paper, delayed/manual simulation and actual manual prop results without combining their win rates.

What this stage proves

Manual prop performance tests signal usefulness, account-rule fit and human execution practicality. It does not prove autonomous broker execution. Measure alert delivery latency, reaction time, missed opportunities, slippage and exit compliance. Use measured manual delays to stress the later autonomous execution model, but keep the execution mechanisms distinct.

Route boundaries

No browser automation, macros, hotkeys, prefilled order tickets, signal-triggered order modification or trade endpoint calls in PROP_MANUAL_ALERTS. Read-only reconciliation must have separately scoped permissions. A deployment smoke test must prove that no prop trading route can be reached.

Apex evaluation/PA activity may be simulated even when payouts are available; do not label every prop fill a live exchange execution. Personal-live deployment remains a separate stage [W1]. Check the applicable agreement/profile at activation and after changes; routine paper work continues independently.

Account rules must be versioned

Keep Apex/Tradovate as Frank's intended manual route. The exact account product, size and purchase cohort are still unspecified. Build a configurable rule engine and synthetic fixtures now; do not invent a funded rule profile or buy/reset accounts automatically.

Field group	Required profile content
Identity / provenance	Firm, account product, platform, size, purchase date/cohort, stage, effective date, source URL/document and last verification date.
Loss floor	Static, end-of-day trailing or intraday unrealized trailing; reference equity; dollar allowance; update schedule; floor cap/lock; commissions; equality breach; payout/reset effects.
Trading-day rules	Session timezone and boundary; holidays; required flat time; daily loss treatment; realized/unrealized inclusion; pending orders and overnight restrictions.
Exposure	Per-instrument contract metadata; micro/mini equivalencies; aggregate maximum; scaling eligibility and timing; correlated hedging restrictions.
Evaluation completion	Profit target, minimum days, consistency formula if any, time limits, fees and disqualifying actions.
Funded / payout	Qualifying days and dollar minimums; consistency numerator/denominator/window; minimum balance/safety net; payout caps, split, payout count and inactivity rules.
Automation	Advisory permission, manual order requirement, native protection permissions, broker/API permission and source agreement.

Corrections to the filled form

The eight-day/five-\$50-day rule belongs to legacy payout eligibility [W4]. Legacy negative open P&L uses starting **profit balance**, with special new/low-profit provisions; it is not 30% of the nominal balance. Half-contract scaling and broader correlated-instrument restrictions require exact legacy semantics [W3]. The form's daily dollar-loss limit and actual evaluation target are still unfilled.

Do not blend account generations. Apex identifies pre-March-1-2026 purchases as legacy [W5]. Its current \$50k EOD payout page instead lists five qualifying \$250 days, a best-day share below 50%, safety-net requirements and a finite payout lifecycle [W6]. These examples explain the discrepancy; neither profile is selected for Frank without account identity.

Why the guide's fast-pass sizing is rejected

Topstep is a comparison only. Its published \$50k Combine has a \$2,000 initial loss allowance [W7] and a \$3,000 target; its current consistency page specifies 55% [W8]. A \$1,000 trade risk consumes half that initial allowance. Nominal account percentages do not measure survival capacity. A rapid winning streak cannot substitute for a durable strategy.

Test account-rule fixtures at just below, exactly at and just above each boundary. Cover intraday highs followed by retracement, end-of-day updates, payout deductions, contract scaling, timezone/DST changes and simultaneous pending orders. If profile state is unknown, stop new prop recommendations, not independent paper research.

Evidence that can support promotion

Use one frozen strategy version for each forward run. Preserve a chronological development segment, validation segment and untouched final holdout. Fit thresholds only on development/validation data. Use walk-forward evaluation and purge overlapping outcomes at split boundaries. No random shuffling of adjacent intraday samples.

Evidence gate	Proposed acceptance
Paper start	A working simulator path, correct accounting and essential failure tests. Do not require a profitable record before allowing safe simulated trades.
Initial performance review	At least 100 completed positions over at least 20 trading sessions and four calendar weeks, with fixed rules and no material execution faults. This is a review floor, not statistical proof.
Paper -> manual prop	Observed net win rate $\geq 70\%$; net expectancy > 0 with a predefined 95% lower confidence bound above zero using session-aware resampling; no hard-rule breach; acceptable stress survival; exact manual account profile. Retain an untouched forward segment.
70% or 75% demonstrated claim	Only use this wording if a prespecified uncertainty method supports a lower bound at the claimed level. Meeting the observed-rate operational gate does not establish the same population lower bound.
Manual prop -> personal live	At least two funded payouts as Frank requested; continued net-positive, rule-compliant performance; no silent reduction to a 60% win-rate gate; separate autonomous broker execution qualification. Frank explicitly activates minimum-size live mode.
Regression / retuning	Material changes create a new version and new forward evidence. A hard execution failure blocks promotion; a missed statistical gate means continue paper or revise under a new version, not force higher risk.

How much confidence 100 trades provide

Outcome	Approximate 95% Wilson interval, IID illustration
70 wins / 100	60.4-78.1%
75 wins / 100	65.7-82.5%
300 wins / 400	70.5-79.0%

These calculations assume independent trades; correlated setups reduce effective evidence. At four trades weekly, 100 trades take about 25 weeks and 400 about 100 weeks. Do not promise a four-week finish or manufacture trade frequency. A 75% observation does not by itself prove a 75% underlying win probability.

Measure net mean R, average/median winners and losers, profit factor, win/breakeven/loss counts, fee drag, tail loss, maximum drawdown, time underwater, MAE/MFE and results by setup/regime. Report confidence and sample counts. Evaluate at fixed milestones (100, 200, 400 positions), not daily significance fishing; use a prespecified sequential method if continuous statistical decisions are necessary.

Streaks, drawdown and robustness

The guide's $0.35^6 = 0.1838\%$ is the probability of six losses in one specified independent six-trade block, not the chance of finding a six-loss run over the whole program. Exact finite-sequence run probabilities illustrate the difference:

Assumed win rate	Six-loss run in 100 trades	In 400 trades
65%	10.87%	37.89%
70%	4.77%	18.33%
75%	1.73%	6.99%

Six fixed 1%-of-start losses lose 6%; six losses at 1% of remaining equity lose 5.852% before costs. Neither implies account failure without a defined floor. Drawdown can breach a rule without six consecutive losses. A four-to-six-trade drawdown is not a guaranteed maximum or recovery time.

Required path simulation

- Bootstrap complete sessions or multi-session blocks from out-of-sample outcomes; retain serial dependence, trade counts and intraday MAE/MFE where available. Sweep block lengths and show uncertainty. IID Bernoulli paths are only a baseline comparison.
- Run the exact sizing, daily stop, trailing floor, payout, scaling and session rules on every path. Model changes to the intraday floor during open profit. Do not rely solely on final closed-trade returns.
- Include lower win-rate scenarios, smaller winners, larger stop slippage, clustered losses, missed alerts, delayed fills, poor liquidity and fewer qualifying setups. Costs and risk changes must affect both winners and losers.
- Report pass-before-breach probability, breach probability, unresolved fraction at horizon, median and 90th-percentile time among passers, worst drawdown quantiles and total program cost. Disclose simulation assumptions and random seed.

Proposed stress acceptance, fixed before the final test

Use at least 10,000 paths over 100 positions and over the actual evaluation horizon, if different. Proposed pilot limits: modeled breach probability $\leq 1\%$ under base assumptions and $\leq 5\%$ under the agreed adverse scenario; stress 95th-percentile drawdown must fit within the usable loss capacity after reserve. These are design tolerances, not measured results or guarantees. Report a confidence interval around modeled probabilities.

Adverse scenario minimums: double modeled trading costs; add one adverse tick per side; replay measured latency at its 95th percentile; reduce win probability by five percentage points in scenario analysis; replay the worst observed loss block. Separately test larger outage/gap shocks, where survival may require no trading rather than a tighter stop.

Do not repeatedly test thousands of strategy variants and report only the winner. Keep an experiment registry and untouched holdout. Passing in favorable conditions does not establish all-regime robustness; restrict deployment to the validated operating envelope.

Data contracts and observability

Record	Minimum fields
Run manifest	run_id; mode; code commit; strategy/config hash; data source and version; instrument metadata; account profile; start/end; clock/timezone policy; cost model.
Market event	event_id; exchange/receive/decision timestamps; bid/ask or bar OHLCV; completed-bar flag; sequence/gap flags; contract; rollover handling.
Signal / rejection	signal_id; setup_id/version; side; level references; feature values known at decision; confirmation matrix; entry zone/expiry; proposed stop/target; invalidation and reason codes.
Risk decision	account equity/floor/headroom; daily P&L; reserved/open risk; stop ticks; tick value; cost/slippage budget; requested/approved quantity; each limit result.
Order / fill / position	intent/order/fill IDs; parent signal; timestamps; requested and filled prices/quantity; fees; slippage; protection status; position lifecycle; reconciliation events.
Manual prop action	alert sent/received/acknowledged; Frank's entered/skipped/expired decision; actual fill/exit; actual protection; deviation reason; import source. Never substitute paper fills.
Trade result	initial R; net P&L/R and dollars; realized payoff; MAE/MFE; holding time; exit reason; rule breaks; setup/region; external-account costs reported separately.

Reconciliation invariants

- One signal can produce at most one filled pilot position. Every fill belongs to an order and every order to a risk-approved intent. Position quantity equals net fills.
- Cash + marked position value - accrued costs reconciles to equity under the account definition. Closed trades, open trades and daily totals reconcile after restart.
- An entry blocked by missing confirmation is distinct from blocked by risk, missing data, expired signal, offline operator or disconnected adapter.
- A manual prop entry rejected/skipped by Frank remains in the opportunity ledger; do not omit it when analyzing selection effects.
- Every post-hoc correction is an appended event with provenance. No silent deletion of losses, failed tests, rejected alerts or previous experiment runs.

Daily report

Show executed positions, net P&L, costs, observed win rate, mean R, maximum intraday drawdown, headroom, rule breaches, data/execution faults, skipped reasons and evidence progress. Include a clear SOURCE ENGINE BENCHMARK / CURRENT BOT RESULT / TARGET comparison. Highlight whether the session had no valid setups or a broken execution path.

Maintain separately: autonomous paper ledger, delayed-manual simulation ledger, actual manual prop ledger and future personal-live ledger. Reuse decision logic and IDs across them, but never pool their outcomes into a misleading single win rate.

Implementation order and done criteria

Priority / deliverable	Done when
P0 - inspect current checkout	Identify active branch, audit-referenced files, available data providers and current startup. Preserve uncommitted work and archive experiments logically rather than deleting them. Produce a short gap list mapped to this brief.
P0 - explicit modes	PAPER_AUTO, PROP_MANUAL_ALERTS and disabled LIVE_AUTO are distinct. Prove paper cannot route real orders and prop mode cannot reach order submission or UI control.
P0 - paper vertical slice	One valid synthetic market sequence produces a signal, correctly sized simulated entry, protective exit and reconciled ledger record. An invalid sequence is skipped with the correct reason. Mark synthetic evidence as synthetic.
P1 - strategy formalization	Implement pages 7-8 using existing validated definitions where available. Add setup-specific tests, causal multi-timeframe alignment, delayed swings and target/stop semantics.
P1 - sizing and controls	Integer sizing, costs, capacity reservations, daily stop, exit management and restart/kill-switch tests pass. Include gap, partial-fill and rejected-protection scenarios.
P1 - full paper sessions	Run during scheduled windows for complete sessions; capture authentic outcomes and no-trade reasons. Report results rather than numbers of audits completed.
P2 - manual alerts / reconciliation	Alert lifecycle, expiry, native-stop instructions and actual fill capture work. Shadow comparison separates timing effects from signal quality. No execution permission is requested.
P2 - exact prop profile	Implement selected account rules only after account identity is known. Boundary fixtures reproduce provided account examples and stage transitions.
P2 - validation report	Frozen out-of-sample run meets or misses fixed gates transparently; clustered path simulation and cost/latency stress are reproducible.
P3 - future personal-live adapter	After promotion: sandbox/qualification evidence for idempotency, protection, reconciliation and failures. Minimum size, restricted account and explicit activation; no automatic stage promotion.

Required implementation tests, not thousands of ceremonial checks

Test real failure modes: future-bar leakage; identical inputs producing identical signals; one decision creating duplicate orders; two pending entries overspending shared capacity; floor ratchet on unrealized profit; stop/target same-bar ambiguity; manual stale alerts; restart with open position; malformed tick metadata; missing calendar; holiday/DST cutoff; delayed fill after cancellation; rejected exit/protection.

Run existing relevant tests and classify failures. Do not add tests that only mirror code or broaden tooling without a concrete need. Keep BTC, neural-network experiments, extra markets, aggressive sizing and VPS/Docker expansion parked unless needed for the selected path.

The audit's statement that the remaining work is small is not a delivery estimate. Report the actual integration gap after inspection. Success for the first milestone is an operating paper path, not an immediate 70% performance claim.

Defaults and remaining decisions

Item	Proposed initial setting / source status
Market / mode	MES / PAPER_AUTO. MES is in Frank's requested universe; selecting it first is a proposed scope reduction.
Strategy	REVERSAL_R1 and CONTINUATION_C1 separately tagged, shared one-position cap; definitions on pages 7-8 are pilot interpretations where repository definitions are absent.
Risk	User ceilings: 0.25-0.5% nominal per trade, 1% nominal daily and 25% of allowed drawdown daily. Stricter paper: \$50k / \$48.5k floor; \$100 reserve; min(\$100, 5% usable capacity, other limits); one MES; \$150 net daily stop; three entries/session.
Payoff / management	Revised answer: ordinary stop or target exit; minimum proposed 1.0 gross target/stop; no partials/trailing. Proposed 30-minute and session/event exits are separately logged safety controls.
Hours / news	09:30-11:30 New York cash-open pilot, flat by 11:30; no premarket entry; major event -10/+15 minute blackout and flat at -5 minutes.
Regime / data	Accumulation/compression candidate and breakout transition recorded causally; two-second quote freshness; two-tick MES spread; 60-second entry intent; 15-minute setup expiry.
Evidence	100 positions + 20 sessions + 4 weeks minimum review; observed net wins >=70% for operational promotion plus expectancy, survival and execution gates.
Risk escalation	Disabled, including guide geometric and fast-pass models. No automated account reset/purchase, withdrawal or live activation.

Resolve before the dependent feature is activated

Decision	What can proceed now
Exact futures feed, fees and history	Reuse the configured accessible futures feed; use conservative labeled simulated costs meanwhile. Synthetic/replay wiring can proceed without credentials.
Apex size/product/cohort and rules	Build rule interfaces, synthetic tests, alerts and paper sessions. Account-specific compliance status stays UNCONFIGURED.
Original engine ledger / ratio notation	Keep 69.9% as the source benchmark and both ratio interpretations; measure Frank's bot independently. Do not treat missing ledger as proof the source is unsuccessful.
Ambiguous strategy/session definitions	Use proposed paper defaults only where absent; present a compact difference list from existing code. Preserve a configurable version for later refinement.
Personal-live broker and activation	No dependency for paper or manual prop alerts. Keep real-money execution disabled until the later stage.

These proposed defaults are implementation scaffolding, not optimized investment advice or recovered source-engine settings. Claude must put them in configuration with provenance labels. Changing them changes the strategy version; already-collected results remain attached to the old version.

Instructions for Claude Code

Continue Frank's existing bot. Deliver the smallest complete autonomous paper-trading path, then the manual prop-alert workflow.

Context to preserve: The source audit describes Frank's current bot. The guide photographs describe a different live signal engine that Frank identifies as proven, with about 70% wins. Use its useful risk principles and reported benchmarks without assigning its results to this implementation. Frank's final goal is fully autonomous personal live trading; prop testing is human-executed.

- **Inspect before editing:** read project instructions, identify the active checkout and inspect the audit-named files. Treat the audit as the supplied baseline, not as authority to overwrite current work. No destructive branch reset or wholesale rewrite.
- **Preserve working components:** reuse the risk engine, kill switch, simulator and ledger. Replace the permanent paper-off path with explicit mode routing; connect actual strategy outputs through existing risk checks.
- **Implement precise rules:** distinguish reversal, continuation and accumulation/compression states, causal timeframes, structural stops, payoff constraints, expiry and regime filters. Existing sound definitions take precedence over proposed defaults; document any discrepancy.
- **Make one path work:** validate synthetic end-to-end entry/exit and accounting, then run genuine full-session paper observations. Do not require proven profitability before beginning paper data collection.
- **Enforce execution separation:** prop mode emits recommendations only. It cannot place, modify, cancel or prefill orders. Frank performs those actions. Personal-live routing remains disabled.
- **Keep evidence honest:** tag synthetic, historical, out-of-sample paper, manual prop and personal-live data separately. Track all trials and skipped signals. Never relabel 69.9% source performance as our measured win rate.
- **Avoid scope drift:** no new markets, neural networks, social/news prediction systems, aggressive sizing variants or replacement platforms without a concrete need and a deliberate scope decision.
- **Handle missing inputs locally:** continue independent paper work; mark only the dependent feature unavailable. Ask Frank concise questions when exact account identity, credentials or a material strategy choice is required. Do not repeat answered questions.
- **Report completion concretely:** list files changed, behavior added, meaningful test results, startup command, configuration, current evidence counts and unresolved blockers. Never claim the bot is profitable or production-ready before the specified evidence exists.

Deliverables from the first implementation milestone

A documented paper startup; one configured MES pilot; causal setup decisions; risk-capped simulated entries and exits; durable ledger and daily report; restart/kill-switch evidence; and a concise report of why authentic setups traded or did not trade. Include an advisory-only alert interface, even if actual prop rules remain unconfigured.

Do not execute historical instructions in the source materials when they conflict with this consolidated brief. In particular, do not import the old blanket no-new-safeguards rule, 60% live gate, nominal 2% fast-pass sizing or automatic prop execution.

Sources and audit trail

[A] Trading Bot Audit - Findings & Path to Paper Trading (1).pdf, revised 11-page copy supplied 29 September 2026. Changed answers on pages 8-10 supersede the older form; exact account rules remain to be verified. Engineering findings were not independently rerun for this document. The original audit is a source reference and is not appended in this standalone edition.

Photo reference	Source filename / content
P1	image0 (1).jpeg - profile and expectancy
P2	image2.jpeg - fixed risk, streaks and geometric model
P3	image4.jpeg - two-phase timing and futures introduction
P4	image5.jpeg - passing/funded risk and comparison
P5	IMG_7487.jpeg - comparison and market conditions
P6	IMG_7488.jpeg - sizing and emotional expectations
P7	IMG_7489.jpeg - assumptions and conclusion

[I1] Frank's 29 September 2026 accumulation/sideways-market screenshot: moving-average slope clutter filter and lower-liquidity session avoidance are candidate research ideas, not empirical results.

Live-engine provenance: Frank's conversation clarification. The uploaded photos are summary evidence, not a trade-level ledger. Source assumptions and results are distinguished throughout. All arithmetic, run probabilities and intervals in this brief were independently recalculated; no new market backtest was run.

Primary references (clickable)

[W1] [Apex User Agreement](#) - manual order control, permitted alerts, agreement precedence.

[W2] [CME Micro E-mini Equity Index Futures FAQ](#) - MES/MNQ contract values.

[W3] [Apex Legacy PA Compliance](#) - legacy negative-P&L, scaling and risk rules.

[W4] [Apex Legacy PA Payout Parameters](#) - legacy payout days and conditions.

[W5] [Apex Legacy Products Overview](#) - account generations.

[W6] [Apex EOD Payouts](#) - current EOD example, not a selected account.

[W7] [Topstep Maximum Loss Limit](#) - comparison of loss capacity.

[W8] [Topstep Consistency](#) - target/consistency example.

[W9] [Forex Factory Calendar](#) - event-calendar context.

[W10] [NYSE Trading Information](#) - 09:30 ET US equity core cash open.

Online terms are account- and date-dependent. Source pages may contain historical summaries; use the applicable agreement and exact product rules when activating a profile. Photos and the original audit are identified here for traceability but are not appended. Their statements are not all adopted requirements.